MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, FIELD EFFECT
DMOS, QUAD N-CHANNEL
TYPES 2N7116, 2N7117, and 2N7118,
JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for an N-channel, lateral DMOS field effect transistor quad for use in high speed switching and multiplexing. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1, side braze dual-in-line, 16 lead.

1.3 Maximum ratings.

Type	P_T at T_C = +25°C 1/	I_D	V_{DS}	V_{SB}	V_{DB}	V_{SD}	V_{GS}	V_{GB}	V_{GD}	Top and T_{STG}
	Package 640 mW, each device 300 mW									
2N7116		50 mA	+20 V	+25 V	+25 V	+20 V	30/-25 V	30/-0.3 V	30/-25 V	-55°C to +150°C
2N7117			+10 V	+15 V	+15 V	+10 V	25/-15 V	25/-0.3 V	25/-15 V	
2N7118			+15 V	+22.5 V	+22.5 V	+15 V	30/-22.5 V	30/0.3 V	30/-22.5 V	

1/ Derate linearly at 6.4 mW/°C above +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Rome Air Development Center, ATTN: RBE-2, Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics at T_C = +25°C.

Type	$V_{(BR)DS}$ $V_{BS} = V_{GS} = -5$ V $I_S = 10$ nA dc	$V_{GS(th)}$ $V_{DS} = V_{GS} = V_T$ $I_D = 1.0$ µA $V_{SB} = 0$ V		$r_{DS(on)}$ $I_D = 1$ mA $V_{SB} = 0$ V $V_{GS} = 5$ V
	V dc	V dc		Ohms
	Min	Min	Max	Max
2N7116	20	0.1	2.0	70
2N7117	10	0.1	2.0	
2N7118	15	0.1	2.0	

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

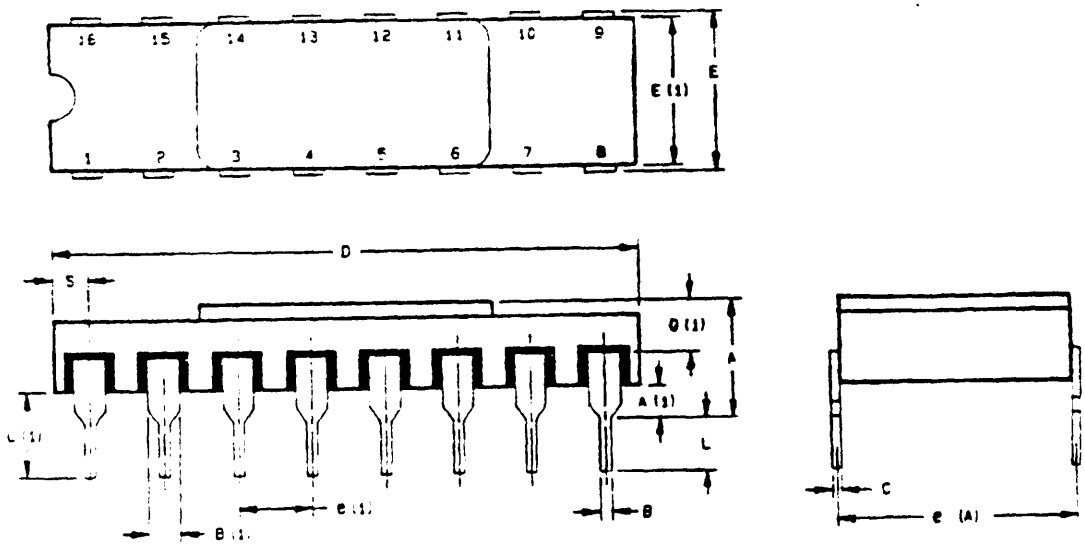
3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

$V_{(BR)DS}$ - Drain to source breakdown voltage, all other terminals as specified.
$I_{SD(off)}$ - Source current through drain (forward biased V_{SD}).
$I_{DS(off)}$ - Drain current through source (forward biased V_{DS}).
$V_{(BR)SD}$ - Source to drain breakdown, all other terminals as specified.
$V_{(BR)DB}$ - Drain to substrate breakdown, all other terminals as specified.
$V_{(BR)SB}$ - Source to substrate breakdown, all other terminals as specified.

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Dimensions				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	.105	.175	2.67	4.44
A(1)	.025	.055	0.64	1.40
B	.015	.021	0.38	0.53
B (1)	.038	.060	0.97	1.52
C	.008	.012	0.20	0.30
D	.770	.830	19.56	21.08
E	.290	.325	7.37	8.26

Dimensions				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
E (1)	.280	.310	7.11	7.87
e (A)	.300 BSC		7.62 BSC	
L	.125	.175	3.18	4.44
L (1)	---	---	---	---
Q (1)	.010	---	0.25	---
S	.020	.065	0.51	1.65
e (1)	.100 BSC		2.54 BSC	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Physical dimensions.

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3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or alloy 52. A copper core is permitted. Lead finish shall be gold, tin plated, or solder dipped. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection (see 6.3). ESD device failures in groups B and C are not to be counted as failures against the lot when verified by the following procedure. ESD failures electrical characteristics:

a. Gate to source body or drain short (primary failure mode).

b. $r_{DS(on)}$ in excess of 100 ohms at V_{GS} = 5 V, I_D = 1 mA (secondary failure mode).

c. I_{fs} below 1 ms (secondary failure mode).

d. $V_{GS(th)}$ shift down (secondary failure mode).

3.5.1 ESD visual characteristics confirming:

a. Small dark short on edge of gate.

b. Crater and gate oxide when viewed under SEM.

Should more than two failures exhibit the ESD failure criteria at a given step in the subgroup testing, then additional in-depth failure analysis is required to verify that ATE testing glitches or defective oxides are not causing the failures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

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Screen (see table II of MIL-STD-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	V_T, $r_{DS(on)}$, $I_{DS(off)1}$, $I_{DS(off)2}$, $I_{DS(off)3}$	V_T, $r_{DS(on)}$, $I_{DS(off)1}$, $I_{DS(off)2}$, $I_{DS(off)3}$
10	Method 1042, test condition B, see 4.3.1.	Method 1042, test condition B.
11	V_T, $r_{DS(on)}$, $I_{DS(off)1}$, $I_{DS(off)2}$, $I_{DS(off)3}$ $\Delta I_{DS(off)}$ = ±2 nA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)}$ = ±10 percent of initial value. 1/ ΔV_T = 0.2 V or ±10 percent of initial value, whichever is greater.	$r_{DS(on)}$, V_T, $I_{DS(off)}$
12	Method 1042, test conditions A and C, see 4.3.1.	Method 1042, test condition A.
13	Subgroups 2 and 3 of table I herein. $\Delta I_{DS(off)}$ = ±2 nA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±10 percent of initial value. 1/ ΔV_T = 0.2 V or ±10 percent of initial value, whichever is greater.	Subgroup 2 of table I herein. $\Delta I_{DS(off)}$ = ±2 nA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±10 percent of initial value. 1/ ΔV_T = 0.2 V or ±10 percent of initial value, whichever is greater.

1/ $\Delta r_{DS(on)}$ is not a measured value, $\Delta r_{DS(on)} = V_{DS(on)}/ID$.

4.3.1 Burn-in. Power burn-in conditions are as follows: T_A = +25°C; V_{DS} = 10 V dc minimum, V_{GS} is such that to guarantee a junction temperature (T_J) of +150°C (+0°C, -15°C), for condition C, t = 240 hours.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for quality conformance inspection in accordance with figure 2 of MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Leakage current, drain to source	3413	$V_{GS} = V_{BS} = -5$ V			10	nA
2N7116		$V_{DS} = 20$ V	$I_{DS(off)1}$			
2N7117		$V_{DS} = 10$ V	$I_{DS(off)2}$			
2N7118		$V_{DS} = 15$ V	$I_{DS(off)3}$			
Leakage current, source to drain	3413	$V_{GD} = V_{BD} = -5$ V			10	nA
2N7116		$V_{DS} = 20$ V	$I_{SD(off)1}$			
2N7117		$V_{DS} = 10$ V	$I_{SD(off)2}$			
2N7118		$V_{DS} = 15$ V	$I_{SD(off)3}$			
Leakage current gate	3411					
2N7116		$V_{DB} = V_{SB} = 0$ V, $V_{GB} = +30$ V	I_{GBS1}		1.0	µA
2N7117		$V_{DB} = V_{SB} = 0$ V, $V_{GB} = +25$ V	I_{GBS2}		1.0	µA
2N7118		$V_{DB} = V_{SB} = 0$ V, $V_{GB} = +30$ V	I_{GBS3}		1.0	µA
Threshold voltage	3403	$V_{DS} = V_{GS} = V_T$, $I_S = 1$ µA, $V_{SB} = 0$ V	V_{T1}	0.1	2.0	V
Drain to source resistance	3421	$I_D = 1.0$ mA, $V_{SB} = 0$ V, $V_{GS} = 5$ V	$r_{DS(on)1}$		70	Ω
		$V_{GS} = 10$ V	$r_{DS(on)2}$		50	Ω
Breakdown voltage, drain to source	3401	$V_{GS} = V_{BS} = -5$ V, $I_D = 1$ µA	$V_{(BR)SD}$			
2N7116				20		V
2N7117				10		V
2N7118				15		V
Breakdown voltage, source to drain	3401	$V_{GD} = V_{BD} = -5$ V, $I_D = 1$ µA	$V_{(BR)SD}$			
2N7116				20		V
2N7117				10		V
2N7118				15		V

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Breakdown voltage, drain to substrate	3401	V_{GB} = 0 V, source open, I_D = 1 µA	$V_{(BR)DB}$			
2N7116				25		V
2N7117				15		V
2N7118				22.5		V
Breakdown voltage, source to substrate	3401	V_{GB} = 0 V, drain open, I_D = 1 µA	$V_{(BR)SB}$			
2N7116				25		V
2N7117				15		V
2N7118				22.5		V
Resistance match	3421	I_D = 1 mA, V_{SB} = 0 V, V_{GS} = 5 V	$r_{DS(match)}$			
2N7116					5	Ω
2N7117					5	Ω
2N7118					5	Ω
Subgroup 3						
High temperature operation:		T_A = +125°C				
Leakage current, drain to source	3413	V_{GS} = V_{BS} = -5 V				
2N7116		V_{DS} = 20 V	$I_{DS(off)4}$		10	µA
2N7117		V_{DS} = 10 V	$I_{DS(off)5}$		5	µA
2N7118		V_{DS} = 15 V	$I_{DS(off)6}$		5	µA
Leakage current, source to drain	3413	V_{GD} = V_{BD} = -5 V, V_{SD} = 10 V	$I_{SD(off)}$		5	µA
Leakage current, gate	3411	V_{GD} = V_{SB} = 0 V	I_{GBS}		100	µA
2N7116		V_{GB} = +25 V				
2N7117		V_{GB} = +25 V				
2N7118		V_{GB} = +30 V				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Threshold voltage	3403	$V_{DS} = V_{GS} = V_T$ $I_S = 1$ µA $V_{SB} = 0$ V	V_{T2}	0.1	2.25	V
Drain to source resistance	3423	$I_D = 1.0$ mA $V_{SB} = 0$ V				
		$V_{GS} = 5$ V	$r_{DS(on)3}$		150	Ω
		$V_{GS} = 10$ V	$r_{DS(on)4}$		100	Ω
Low temperature operation:		$T_A = -55°C$				
Threshold voltage	3403	$V_{DS} = V_{GS} = V_T$ $I_S = 1$ µA $V_{SB} = 0$ V	V_{T3}	0.1	2.25	V
Drain to source resistance	3423	$I_D = 1.0$ mA $V_{SB} = 0$ V				
		$V_{GS} = 5$ V	$r_{DS(on)5}$		70	Ω
		$V_{GS} = 10$ V	$r_{DS(on)6}$		50	Ω
Subgroup 4						
Forward transconductance		$V_{DS} = 10$ V $V_{SB} = 0$ V $I_D = 20$ mA $f = 1$ kHz	gfs	10		ns
Capacitance:		$V_{DS} = 10$ V, $f = 1$ MHz $V_{GS} = V_{BS} = -15$ V				
Small signal capacitance gate node			C(GS+GD+GB)		3.5	pF
Small signal capacitance drain node			C(GD+DB)		1.5	pF
Small signal capacitance source node			C(GS+SB)		6.0	pF
Small signal capacitance reverse transfer			CDG		0.5	pF

1/ For sampling plan, see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1.
Subgroup 2		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 3		
Thermal shock (temperature cycling)	1051	
Hermetic seal	1071	
a. Fine leak b. Gross leak		
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7.
Decap internal visual (design verification)	2075	
SEM (when specified)	2077	
Bond strength	2037	Test condition A. All internal wires for each device shall be pulled separately.
Die shear	2017	
Subgroup 4		
Intermittent operation life	1037	Condition D, power-on time shall be 3 minutes minimum, 2,000 cycles.
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, 7, 8, and 9.

See footnote at end of table.

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TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 5		
Accelerated steady-state operation life	1042	Condition C, 120 hours, V_{DS} = 10 V dc, P_T = 1.2 W at T_A = +75°C or adjusted as required by the chosen T_A to give an average lot T_J = +150°C. Marking legibility requirements shall not apply.
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, 7, 8, and 9.
Bond strength (Al-Au die interconnects only)	2037	Test condition A
Subgroup 6		
Not applicable		

1/ For sampling plan, see MIL-S-19500.

TABLE IIb. Group B inspection for JANTX and JANTXV.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 2		
Thermal shock (temperature cycling)	1051	

1/ See footnote at end of table.

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TABLE IIb. Group B inspection for JANTX and JANTXV - Continued.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 2 - Continued		
Hermetic seal	1071	
a. Fine leak b. Gross leak		
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7.
Subgroup 3		
Steady-state operation life	1042	Test condition A, V_{GS} = 80 percent of rated breakdown, V_{DS} = 0 V, T_A = +125°C, t = 340 hours (option: T_A = +150°C, t = 170 hours).
Electrical measurements		See table V, steps 1, 4, and 5.
Subgroup 4		
Decap internal visual (design verification)	2075	
Bond strength	2037	Test condition A; all internal bond wires for each device shall be pulled separately.
Subgroup 5		
Not applicable		
Subgroup 6		
High-temperature life (nonoperating)	1032	T_A = +150°C
Electrical measurements		See table V, steps 1, 4, and 5.

1/ For sampling plan, see MIL-S-19500.

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TABLE III. Group C inspection.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1.
Subgroup 2		
Thermal shock (glass strain)	1056	Test condition A
Terminal strength (lead fatigue)	2036	Test condition E
Hermetic seal	1071	
a. Fine leak b. Gross leak		
Moisture resistance	1021	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, 7, 8, and 9.
Subgroup 3		
Shock	2016	
Vibration, variable frequency	2056	
Constant acceleration	2006	20,000 g's, minimum
Electrical measurements		See table V, steps 1, 4, and 5.
Subgroup 4		
Salt atmosphere (corrosion)	1041	
Subgroup 5		
Not applicable		
Subgroup 6		
Steady-state operation life (LTPD)	1039	Test condition A, V_{DS} = 80 percent of rated breakdown, V_{GS} = 0 V, T_A = +125°C, t = 1,000 hours (Option: T_A = +150°C, t = 500 hours)
Electrical measurements		See table V, steps 1, 4, and 5.

1/ For sampling plan, see MIL-S-19500.

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TABLE IV. Group E inspection (all quality levels) for qualification only.

Inspection 1/	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			45 devices c = 0
Thermal shock (glass strain)	1051	-55°C to 150°C; 200 cycles	
Hermetic seal	1071		
a. Fine leak b. Gross leak			
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, 7, and 8.	
Subgroup 2			45 devices c = 0
Steady-state reverse life	1042	Test condition A, 1,000 hours	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, 7, 8, and 9.	
Steady-state gate bias	1042	Test condition B, 1,000 hours	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, 7, 8, and 9.	
Subgroup 3			5 devices c = 0
Destructive physical analysis		Photos of cross sections shall be submitted in the qualification report. Vendor shall retain duplicate photos.	
Subgroup 4			5 devices c = 0
Thermal resistance	3131		
Subgroup 5			
N/A			
Subgroup 6			10 devices c = 0
ESD	1020		

1/ A separate sample may be pulled for each test.

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TABLE V. Groups A, B, and C electrical end-point measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Leakage current, drain to source	3413	$V_{GS} = V_{BS} = -5$ V			10	nA
	2N7116		$V_{DS} = 20$ V	$I_{DS(off)1}$			
	2N7117		$V_{DS} = 10$ V	$I_{DS(off)2}$			
	2N7118		$V_{DS} = 15$ V	$I_{DS(off)3}$			
2	Leakage current, source to drain	3413	$V_{GD} = V_{BD} = -5$ V			10	nA
	2N7116		$V_{SD} = 20$ V	$I_{SD(off)1}$			
	2N7117		$V_{SD} = 10$ V	$I_{SD(off)2}$			
	2N7118		$V_{SD} = 15$ V	$I_{SD(off)3}$			
3	Leakage current gate	3411	$V_{GD} = V_{SB} = 0$ V				µA
	2N7116		$V_{GB} = +30$ V	$I_{(GBS)1}$		1.0	
	2N7117		$V_{GB} = +25$ V	$I_{(GBS)2}$		1.0	
	2N7118		$V_{GB} = +30$ V	$I_{(GBS)3}$		1.0	
4	Threshold voltage	3403	$V_{DS} = V_{GS} = V_T$ $I_S = 1$ µA $V_{SB} = 0$ V	V_{T1}	0.1	2.0	V
	2N7116						
	2N7117						
	2N7118						
5	Drain to source resistance	3423	$I_S = 1.0$ mA $V_{SB} = 0$ V				
			$V_{GS} = 5$ V	$r_{DS(on)1}$		70	Ω
			$V_{SB} = 10$ V	$r_{DS(on)2}$		50	Ω
6	Breakdown voltage, drain to source	3401	$V_{GS} = V_{BS} = -5$ V $I_D = 1$ µA				
	2N7116			$V_{BR(DS)1}$	20		V
	2N7117			$V_{BR(DS)2}$	10		V
	2N7118			$V_{BR(DS)3}$	15		V
7	Breakdown voltage, source to drain	3401	$V_{GD} = V_{BD} = -5$ V $I_D = 1$ µA				
	2N7116			$V_{BR(SD)1}$	20		V
	2N7117			$V_{BR(SD)2}$	10		V
	2N7118			$V_{BR(SD)3}$	15		V
8	Breakdown voltage, drain to substrate	3401	$V_{GB} = 0$ V, source open, $I_D = 1$ µA	$V_{(BR)DB}$			
	2N7116				25		V
	2N7117				15		V
	2N7118				22.5		V
9	Breakdown voltage, source to substrate	3401	$V_{GB} = 0$ V, drain open, $I_D = 1$ µA	$V_{(BR)SB}$			
	2N7116				25		V
	2N7117				15		V
	2N7118				22.5		V

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.4 Group E inspection. Group E inspection shall be in accordance with table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead material and finish may be specified (see 3.3.1).

6.3 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ kΩ, whenever bias voltage is to be applied drain-to-source.

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6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's part number. This information in no way implies that manufacturer's part numbers are suitable as a substitute for the military Part or Identifying Number (PIN).

PIN	Manufacturer's code CAGE	Manufacturer's and user's part number
2N7116	17856	SD5000I
2N7117	17856	SD5001I
2N7118	17856	SD5002I

Vendor CAGE number	Vendor name and address
17856	Siliconix, Incorporated 2201 Laurelwood Road Santa Clara, CA 95054

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, ER, MI
Navy - SH
Air Force - 11, 19, 85, 99
DLA - ES
NASA - NA

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13, 15

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1110)

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